Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir
Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

Abdullah Faqiih Fadhlurrahman - 5024241092

2026

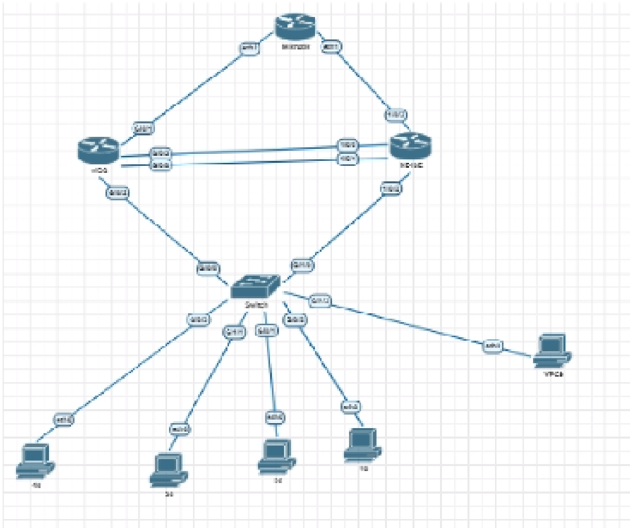
1 Langkah-Langkah Percobaan

1.1 Persiapan Topologi Jaringan

1.1.1 1. Membuat Topologi Jaringan

Langkah pertama yang dilakukan adalah membuat topologi jaringan pada GNS3/PNETLab sesuai dengan topologi yang terdapat pada modul praktikum. Topologi terdiri dari Cisco Switch, Cisco Router, Huawei Router, MikroTik Router, lima buah PC client, serta koneksi antar-router yang akan digunakan untuk implementasi OSPF multivendor.

Selain itu dilakukan penyesuaian interface antar perangkat agar sesuai dengan tabel addressing dan koneksi yang telah ditentukan pada modul.



Gambar 1: Topologi Jaringan Modul 5

1.1.2 2. Menentukan Tabel Addressing

Setelah topologi selesai dibuat, dilakukan perencanaan alamat IP untuk setiap VLAN, client, dan link antar-router sesuai dengan tabel addressing pada modul. Tahap ini bertujuan untuk memastikan tidak terjadi konflik alamat IP selama proses konfigurasi.

⚠ Warning

DISCLAIMER: Ada kesalahan pada dua link jalur cisco-huawei di topologi di atas. Seharusnya:

- `g0/0` terhubung ke `ethernet 1/0/4`
- `g0/3` terhubung ke `ethernet 1/0/0`

1.2 Tabel Addressing Vlan

VLAN ID	Nama VLAN	Network	Gateway	Keterangan
10	FINANCE	192.168.10.0/24	192.168.10.1	DHCP Server
20	HR	192.168.20.0/24	192.168.20.1	DHCP Server
30	IT	192.168.30.0/24	192.168.30.1	Static IP
40	MARKETING	192.168.40.0/24	192.168.40.1	Static IP

1.3 Tabel Addressing client

Perangkat	VLAN	IP Address	Gateway	Keterangan
PC Finance	10	DHCP	192.168.10.1	Mendapat IP otomatis dari DHCP Server
PC HR	20	DHCP	192.168.20.1	Mendapat IP otomatis dari DHCP Server
PC IT	30	192.168.30.10/24	192.168.30.1	IP static
PC Marketing	40	192.168.40.10/24	192.168.40.1	IP static
PC LAN Huawei	-	192.168.50.10/24	192.168.50.1	LAN tambahan pada sisi Huawei

1.4 Tabel Link Antar Router

Link	Network	Perangkat	Interface	IP Address
Cisco ↔ Huawei Link 1	10.10.10.0/30	Cisco Router	Gi0/3	10.10.10.1/30
Cisco ↔ Huawei Link 1	10.10.10.0/30	Huawei Router	Ethernet1/0/0	10.10.10.2/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Cisco Router	Gi0/0	10.10.11.1/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Huawei Router	Ethernet1/0/4	10.10.11.2/30
Cisco ↔ MikroTik	10.10.30.0/30	Cisco Router	Gi0/1	10.10.30.2/30
Cisco ↔ MikroTik	10.10.30.0/30	MikroTik	ether2	10.10.30.1/30
MikroTik ↔ Huawei	10.10.20.0/30	MikroTik	ether1	10.10.20.2/30
MikroTik ↔ Huawei	10.10.20.0/30	Huawei Router	Ethernet1/0/3	10.10.20.1/30

Gambar 2: Tabel Addressing Jaringan

1.2 Praktikum 1 Konfigurasi Cisco Switch

1.2.1 1. Masuk ke Terminal Cisco Switch

Cisco Switch diakses menggunakan terminal kemudian masuk ke mode konfigurasi.

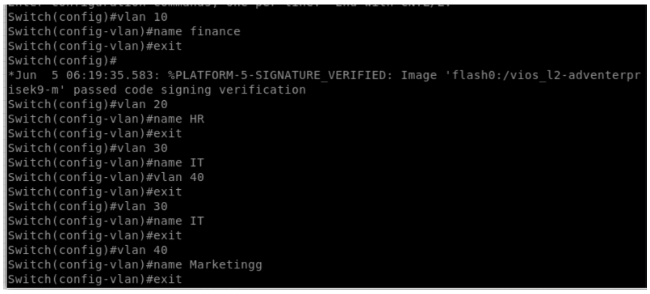
```
enable
configure terminal
```

1.2.2 2. Membuat VLAN

VLAN dibuat sesuai dengan pembagian divisi yang terdapat pada modul.

```
vlan 10
name Finance

vlan 20
name HR

vlan 30
name IT

vlan 40
```

```
name Marketing
```

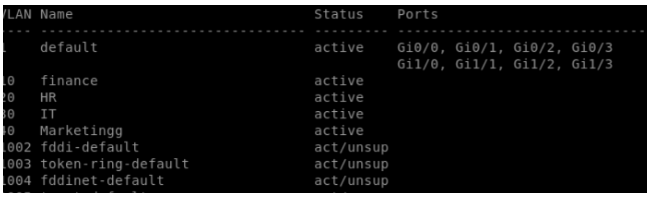


Gambar 3: Pembuatan VLAN

1.2.3 3. Verifikasi VLAN

Verifikasi dilakukan menggunakan perintah berikut.

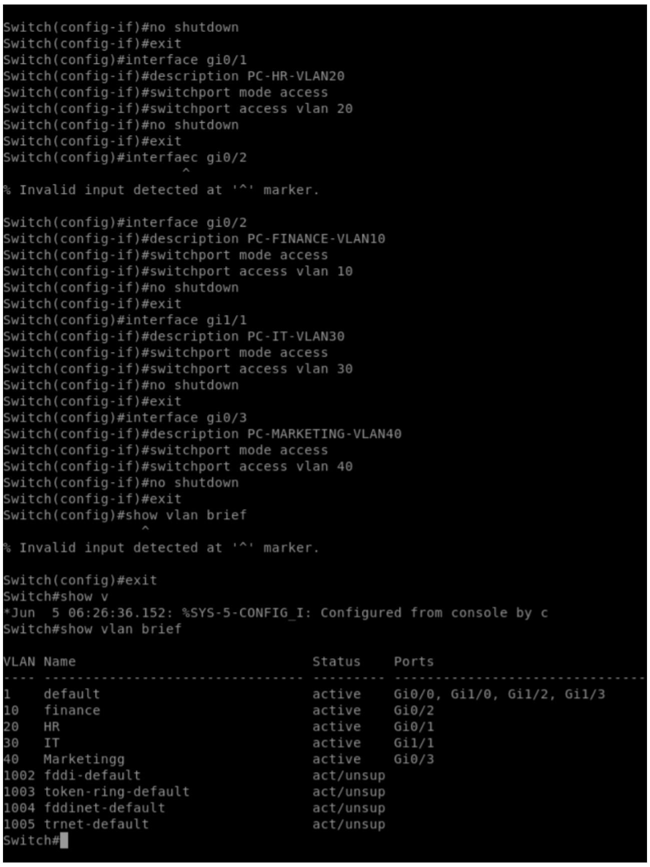
```
show vlan brief
```



Gambar 4: Verifikasi VLAN

1.2.4 4. Konfigurasi Access Port untuk Setiap VLAN

Port switch dikonfigurasi sebagai access port sesuai VLAN masing-masing.

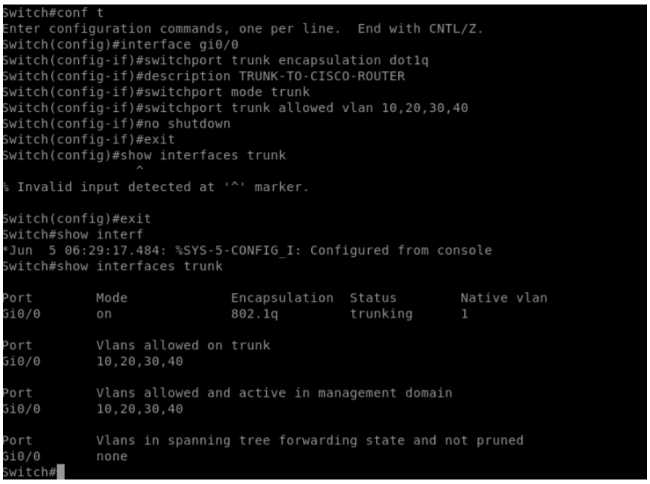


```
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interfaec gi0/2
                       ^
% Invalid input detected at '^' marker.

Switch(config)#interface gi0/2
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi1/1
Switch(config-if)#description PC-IT-VLAN30
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/3
Switch(config-if)#description PC-MARKETING-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#show vlan brief
                  ^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#show v
*Jun  5 06:26:36.152: %SYS-5-CONFIG_I: Configured from console by c
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/0, Gi1/0, Gi1/2, Gi1/3
10   finance                          active    Gi0/2
20   HR                               active    Gi0/1
30   IT                               active    Gi1/1
40   Marketingg                       active    Gi0/3
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

Gambar 5: Konfigurasi Access Port

1.2.5 5. Konfigurasi Trunk ke Cisco Router

Interface yang terhubung ke Cisco Router dikonfigurasi sebagai trunk.



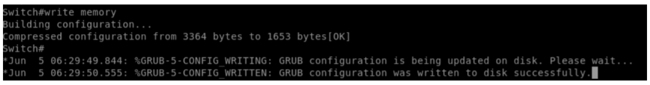
```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
                  ^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#show interf
*Jun  5 06:29:17.484: %SYS-5-CONFIG_I: Configured from console
Switch#show interfaces trunk

Port        Mode             Encapsulation  Status        Native vlan
Gi0/0       on               802.1q         trunking      1

Port        Vlans allowed on trunk
Gi0/0       10,20,30,40

Port        Vlans allowed and active in management domain
Gi0/0       10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gi0/0       none
Switch#
```

Gambar 6: Konfigurasi Trunk

1.2.6 6. Menyimpan Konfigurasi

Konfigurasi yang telah dibuat disimpan ke memori perangkat.

```
write memory
```



Gambar 7: Menyimpan Konfigurasi

1.3 Praktikum 2 Konfigurasi Cisco Router

1.3.1 1. Masuk ke Cisco Router

Cisco Router diakses melalui terminal dan masuk ke mode konfigurasi.

1.3.2 2. Konfigurasi Interface Trunk ke Switch

Interface yang terhubung ke switch diaktifkan untuk mendukung subinterface VLAN.

1.3.3 3. Konfigurasi Subinterface VLAN 10

Subinterface untuk VLAN 10 dibuat dan diberikan alamat gateway.

1.3.4 4. Konfigurasi Subinterface VLAN 20

Subinterface VLAN 20 dikonfigurasi sebagai gateway VLAN HR.

1.3.5 5. Konfigurasi Subinterface VLAN 30

Subinterface VLAN 30 dikonfigurasi sebagai gateway VLAN IT.

1.3.6 6. Konfigurasi Subinterface VLAN 40

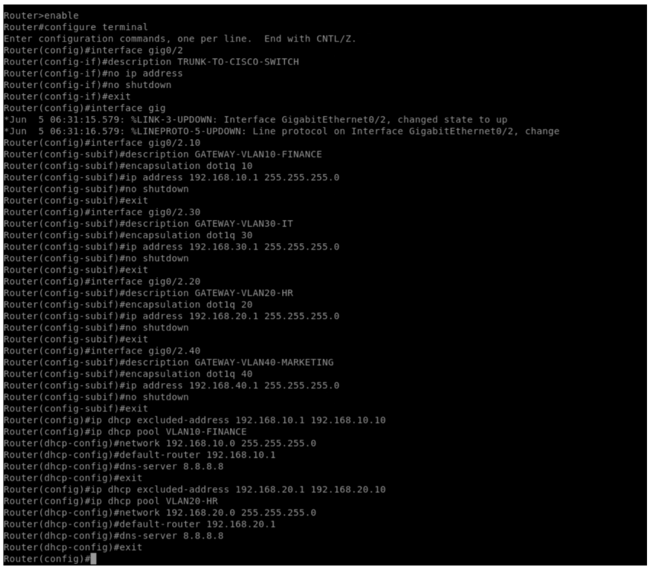
Subinterface VLAN 40 dikonfigurasi sebagai gateway VLAN Marketing.

1.3.7 7. Konfigurasi DHCP Server VLAN 10

DHCP Pool VLAN 10 dibuat agar client memperoleh IP secara otomatis.

1.3.8 8. Konfigurasi DHCP Server VLAN 20

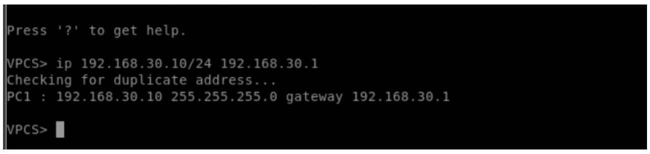
DHCP Pool VLAN 20 dibuat untuk client HR.



Gambar 8: langkah 1-8

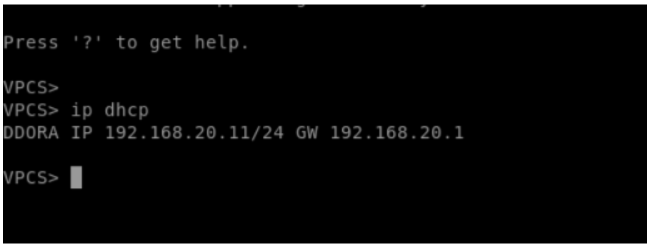
1.3.9 9. Konfigurasi IP Static VLAN 30 dan VLAN 40

Client pada VLAN 30 dan VLAN 40 menggunakan IP statis sesuai tabel addressing.



Gambar 9: Konfigurasi IP Static VLAN 30

Gambar 10: Konfigurasi IP Static VLAN 40



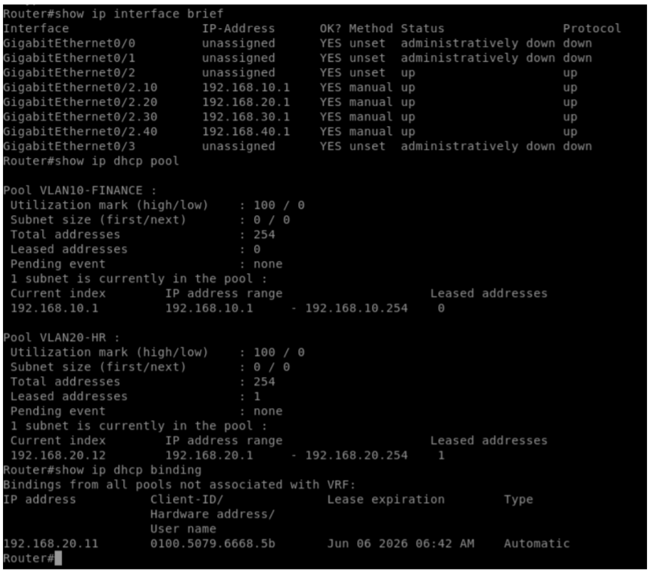
Gambar 11: Konfigurasi IP DHCP

1.3.10 10. Verifikasi Cisco Router

Verifikasi dilakukan menggunakan beberapa perintah monitoring.

```
show ip interface brief
show ip dhcp pool
```

```
show ip dhcp binding
```

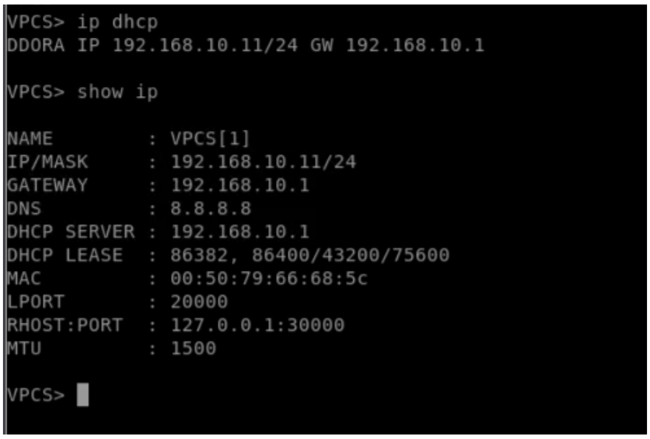


```
Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES unset  administratively down down
GigabitEthernet0/1         unassigned      YES unset  administratively down down
GigabitEthernet0/2         unassigned      YES unset  up                    up
GigabitEthernet0/2.10      192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20      192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30      192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40      192.168.40.1    YES manual up                    up
GigabitEthernet0/3         unassigned      YES unset  administratively down down
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.10.1         192.168.10.1     - 192.168.10.254    0

Pool VLAN20-HR :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.20.12        192.168.20.1     - 192.168.20.254    1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.20.11       0100.5079.6668.5b       Jun 06 2026 06:42 AM    Automatic
Router#
```

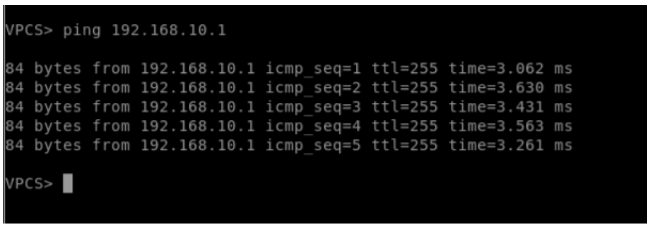
Gambar 12: Verifikasi Cisco Router

1.3.11 11. Pengujian dari VPCS

Pengujian dilakukan dengan melakukan ping ke gateway masing-masing VLAN.

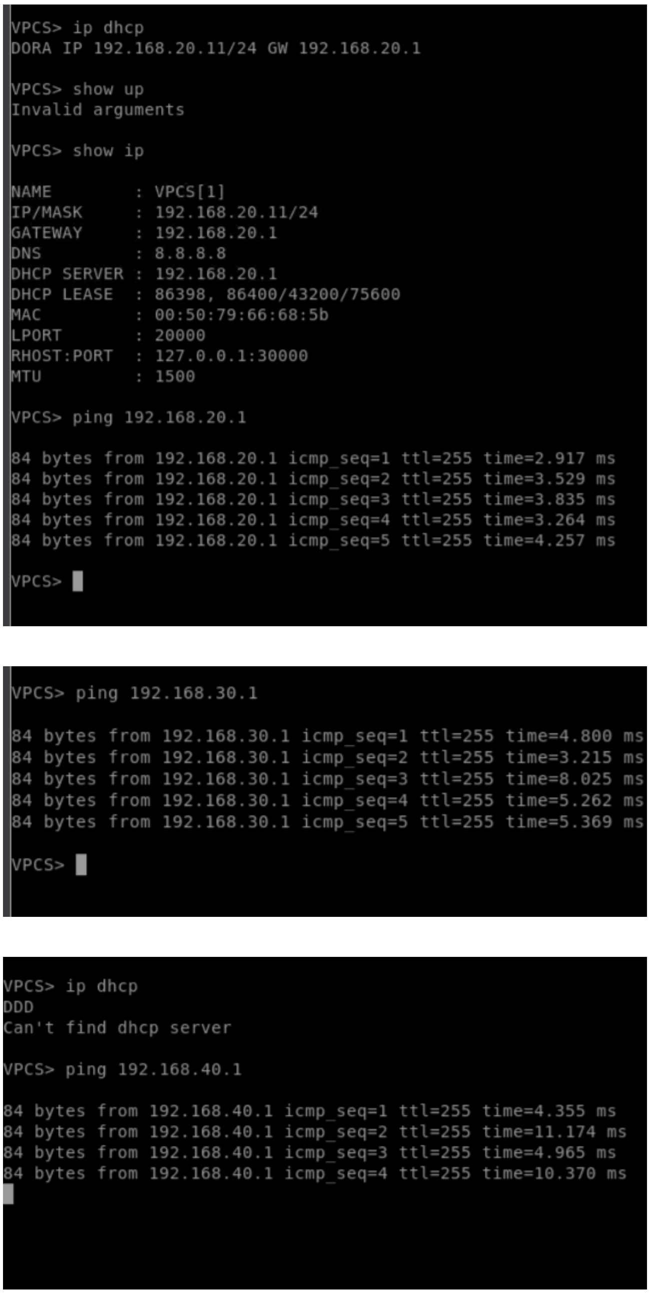


```
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.10.11/24
GATEWAY     : 192.168.10.1
DNS         : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86382, 86400/43200/75600
MAC         : 00:50:79:66:68:5c
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS>
```

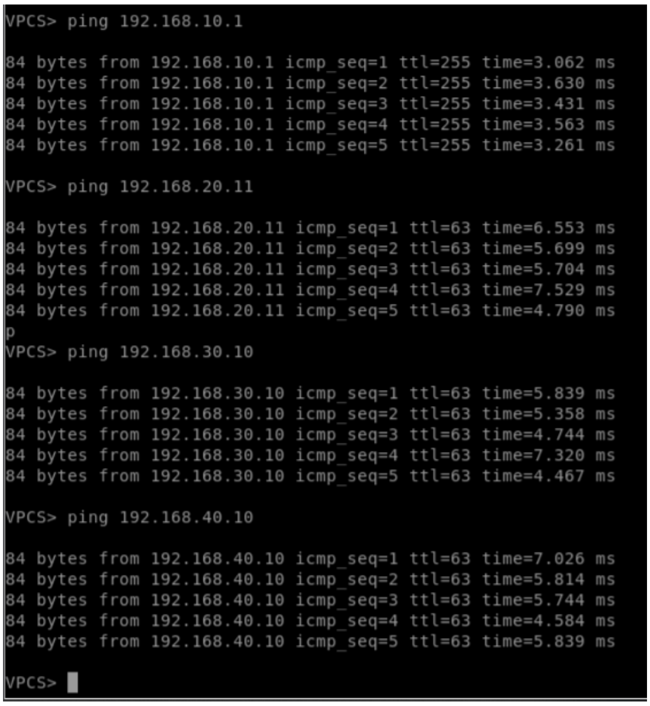


```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.062 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.630 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.431 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.563 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.261 ms

VPCS>
```

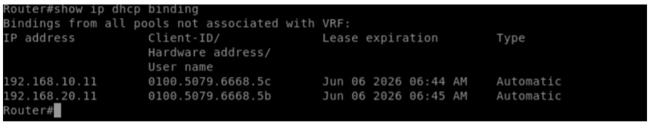
Gambar 13: Pengujian VPCS pada seluruh VLAN



Gambar 14: Pengujian VPCS pada seluruh VLAN



```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.062 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.630 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.431 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.563 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.261 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=6.553 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.699 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.704 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=7.529 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.790 ms
p
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=5.839 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.358 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.744 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=7.320 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=4.467 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=7.026 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.814 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.744 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.584 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=5.839 ms

VPCS>
```



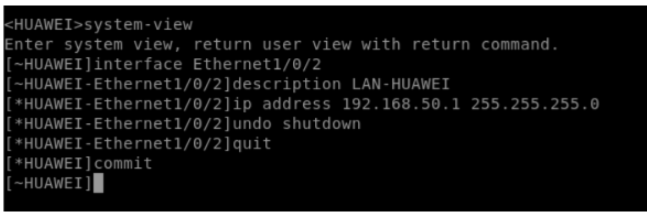
```
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0100.5079.6668.5c       Jun 06 2026 06:44 AM    Automatic
192.168.20.11       0100.5079.6668.5b       Jun 06 2026 06:45 AM    Automatic
Router#
```

Gambar 15: Pengujian VPCS pada seluruh VLAN

1.4 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antar Router

1.4.1 1. Konfigurasi LAN Huawei

Huawei Router dikonfigurasi untuk jaringan LAN Huawei.



```
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 16: Konfigurasi LAN Huawei

1.4.2 2. Konfigurasi IP VPC LAN Huawei

Alamat IP pada VPCS LAN Huawei dikonfigurasi sesuai tabel addressing.

```
VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:5d
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500
```

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=13.408 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.324 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.915 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.182 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.026 ms
```

Gambar 17: Konfigurasi IP VPCS Huawei

1.4.3 3. Konfigurasi IP Link Cisco-Huawei Link 1

```
Router(config)#interface gi0/3
Router(config-if)#description LINK1-TO-HUAWEI
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown exit
                               ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown
Router(config-if)#exit
```

```
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 18: Konfigurasi Link Cisco-Huawei 1

1.4.4 4. Konfigurasi IP Link Cisco-Huawei Link 2

```
Router(config)#interface gi0/0
Router(config-if)#description LINK2-TO-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

```
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4]description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4]undo shutdown
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]commit
```

Gambar 19: Konfigurasi Link Cisco-Huawei 2

1.4.5 5. Konfigurasi IP Link Huawei-MikroTik

```
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3]description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3]undo shutdown
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 20: Konfigurasi Link Huawei-MikroTik

1.4.6 6. Konfigurasi IP Link Cisco-MikroTik

```
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
```

```
[admin@MikroTik] >> /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
```

Gambar 21: Konfigurasi Link Cisco-MikroTik

1.4.7 7. Verifikasi Interface

Seluruh interface diverifikasi agar berada pada status up.

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.11.1      YES manual up                    up
GigabitEthernet0/1     10.10.30.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/2.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/2.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/3     10.10.10.1      YES manual up                    up
Router#
```

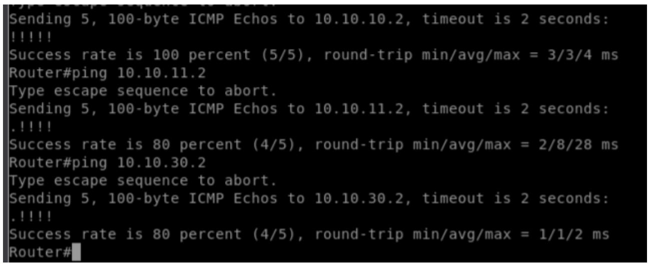
```
[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                         IP Address/Mask      Physical   Protocol VPN
Ethernet1/0/0                     10.10.10.2/30        up         up       --
Ethernet1/0/0.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/1                     unassigned           up         down     --
Ethernet1/0/1.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/2                     192.168.50.1/24      up         up       --
Ethernet1/0/2.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/3                     10.10.20.1/30        up         up       --
Ethernet1/0/3.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/4                     10.10.11.2/30        up         up       --
Ethernet1/0/4.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/5                     unassigned           up         down     --
Ethernet1/0/5.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/6                     unassigned           up         down     --
Ethernet1/0/6.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/7                     unassigned           up         down     --
Ethernet1/0/7.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/8                     unassigned           up         down     --
Ethernet1/0/8.4094                128.1.138.137/16     up         up       l3vpn
Ethernet1/0/9                     unassigned           up         down     --
Ethernet1/0/9.4094                128.1.138.137/16     up         up       l3vpn
GigabitEthernet0/0/0              unassigned           up         down     --
LoopBack2147483647                128.1.138.137/16     up         up(s)    l3vpn
NULL0                             unassigned           up         up(s)    --

[~HUAWEI]
```

```
[admin@MikroTik] >> /ip address print
Flags: X - disabled, I - invalid, D - dynamic
 #   ADDRESS            NETWORK         INTERFACE
 0   ;;; TO-HUAWEI
     10.10.20.2/30      10.10.20.0      ether1
 1   ;;; TO-CISCO
     10.10.30.2/30      10.10.30.0      ether2
```

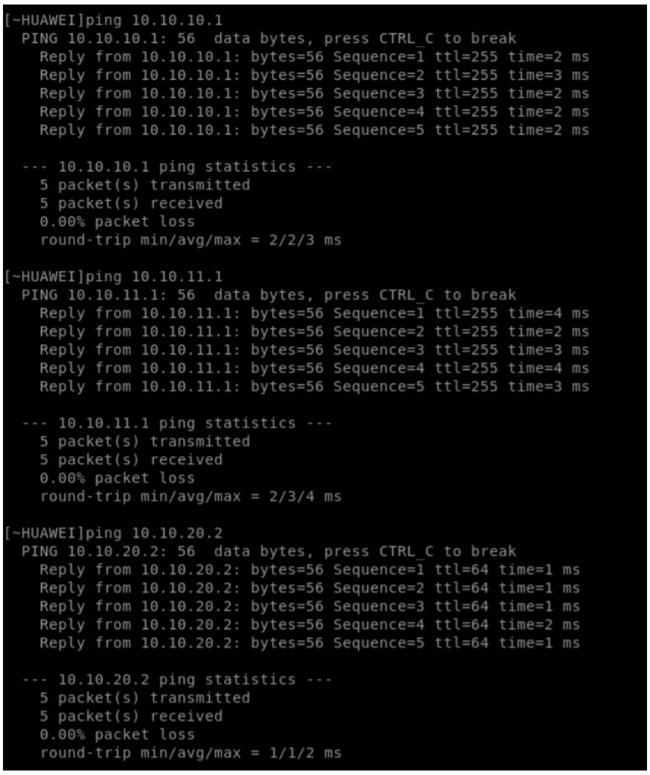
Gambar 22: Verifikasi Interface

1.4.8 8. Verifikasi Ping Antar Router

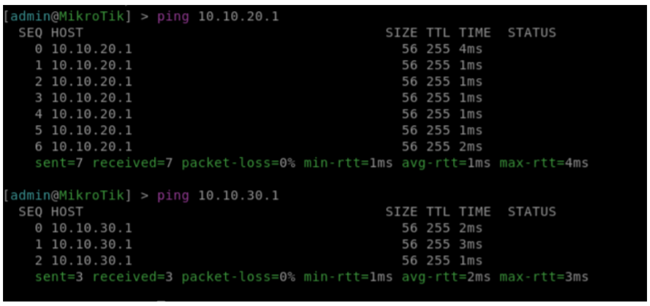
Dilakukan pengujian ping antar router.



```
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/8/28 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#
```



```
[~HUAWEI]ping 10.10.10.1
  PING 10.10.10.1: 56  data bytes, press CTRL_C to break
    Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
    Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

  --- 10.10.10.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
  PING 10.10.11.1: 56  data bytes, press CTRL_C to break
    Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=4 ms
    Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
    Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
    Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=4 ms
    Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

  --- 10.10.11.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/3/4 ms

[~HUAWEI]ping 10.10.20.2
  PING 10.10.20.2: 56  data bytes, press CTRL_C to break
    Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
    Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

  --- 10.10.20.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/2 ms
```

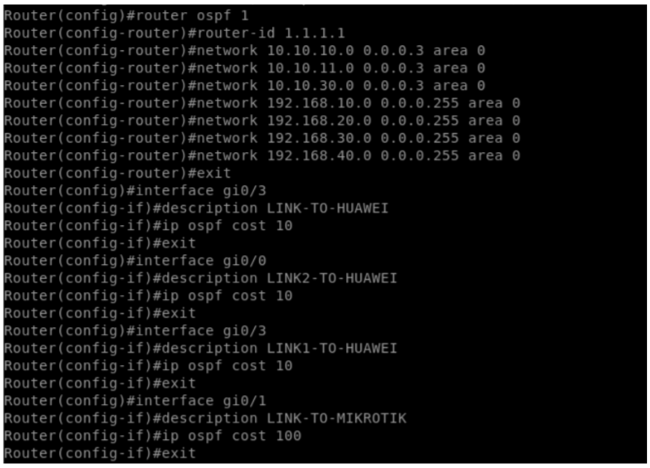


```
[admin@MikroTik] > ping 10.10.20.1
  SEQ HOST                                     SIZE TTL TIME  STATUS
    0 10.10.20.1                                 56 255 4ms
    1 10.10.20.1                                 56 255 1ms
    2 10.10.20.1                                 56 255 1ms
    3 10.10.20.1                                 56 255 1ms
    4 10.10.20.1                                 56 255 1ms
    5 10.10.20.1                                 56 255 1ms
    6 10.10.20.1                                 56 255 2ms
    sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
  SEQ HOST                                     SIZE TTL TIME  STATUS
    0 10.10.30.1                                 56 255 2ms
    1 10.10.30.1                                 56 255 3ms
    2 10.10.30.1                                 56 255 1ms
    sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=3ms
```

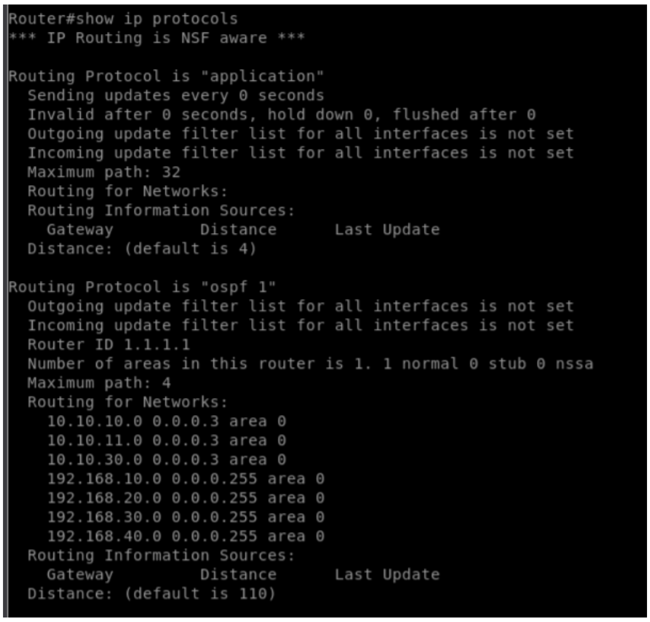
Gambar 23: Verifikasi Ping Antar Router

1.5 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1.5.1 1. Konfigurasi OSPF pada Cisco Router



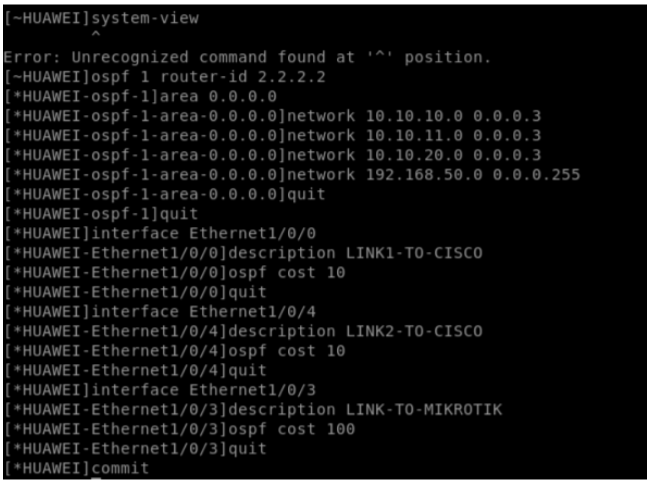
```
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)#description LINK-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#description LINK2-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/3
Router(config-if)#description LINK1-TO-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit
```



```
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)
```

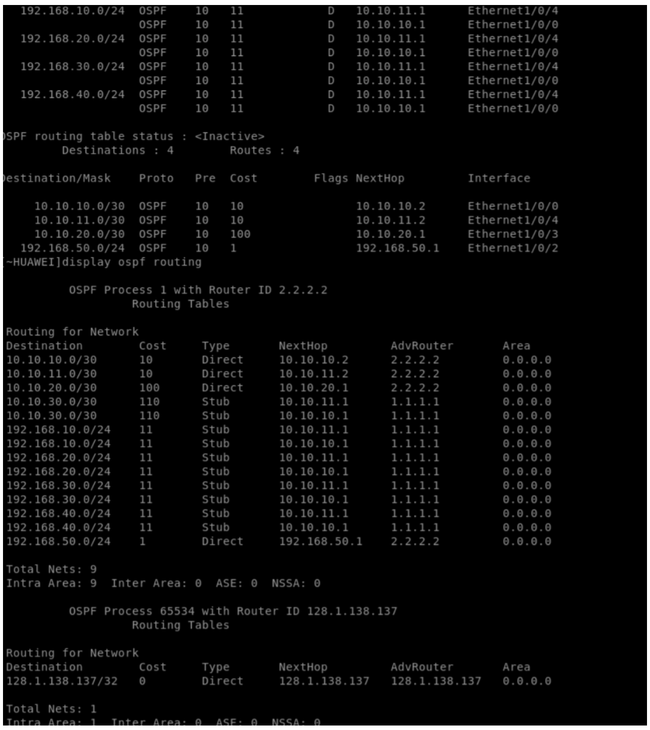
Gambar 24: Konfigurasi OSPF Cisco

1.5.2 2. Konfigurasi OSPF pada Huawei Router



```
[~HUAWEI]system-view
          ^
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-TO-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-TO-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
```

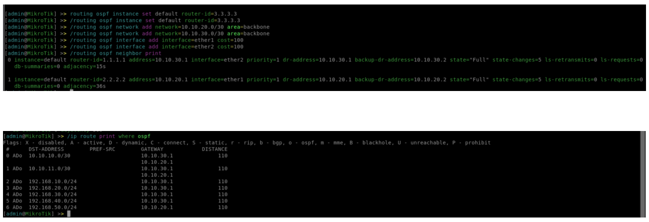
```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor


         OSPF Process 1 with Router ID 2.2.2.2
               Neighbors

 Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.10.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.10.1        BDR: 10.10.10.2        MTU: 1500
   Dead timer due in  36  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m28s
   Neighbor Up Time : 2026-06-05 07:15:57
   Authentication Sequence: [ 0 ]

 Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
 Router ID: 1.1.1.1              Address: 10.10.11.1
   State: Full           Mode:Nbr is Slave      Priority: 1
   DR: 10.10.11.1        BDR: 10.10.11.2        MTU: 1500
   Dead timer due in  32  sec
   Retrans timer interval: 5
   Neighbor is up for 00h00m28s
   Neighbor Up Time : 2026-06-05 07:15:57
   Authentication Sequence: [ 0 ]
```



```
  192.168.10.0/24  OSPF    10   11           D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11           D   10.10.10.1      Ethernet1/0/0
  192.168.20.0/24  OSPF    10   11           D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11           D   10.10.10.1      Ethernet1/0/0
  192.168.30.0/24  OSPF    10   11           D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11           D   10.10.10.1      Ethernet1/0/0
  192.168.40.0/24  OSPF    10   11           D   10.10.11.1      Ethernet1/0/4
                   OSPF    10   11           D   10.10.10.1      Ethernet1/0/0

OSPF routing table status : <Inactive>
        Destinations : 4        Routes : 4

Destination/Mask    Proto   Pre  Cost       Flags NextHop        Interface

    10.10.10.0/30  OSPF    10   10               10.10.10.2      Ethernet1/0/0
    10.10.11.0/30  OSPF    10   10               10.10.11.2      Ethernet1/0/4
    10.10.20.0/30  OSPF    10   100              10.10.20.1      Ethernet1/0/3
  192.168.50.0/24  OSPF    10   1                192.168.50.1    Ethernet1/0/2
[~HUAWEI]display ospf routing

         OSPF Process 1 with Router ID 2.2.2.2
                  Routing Tables

 Routing for Network
 Destination        Cost     Type       NextHop         AdvRouter       Area
 10.10.10.0/30      10       Direct     10.10.10.2      2.2.2.2         0.0.0.0
 10.10.11.0/30      10       Direct     10.10.11.2      2.2.2.2         0.0.0.0
 10.10.20.0/30      100      Direct     10.10.20.1      2.2.2.2         0.0.0.0
 10.10.30.0/30      110      Stub       10.10.11.1      1.1.1.1         0.0.0.0
 10.10.30.0/30      110      Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.10.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.10.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.20.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.20.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.30.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.30.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.40.0/24    11       Stub       10.10.11.1      1.1.1.1         0.0.0.0
 192.168.40.0/24    11       Stub       10.10.10.1      1.1.1.1         0.0.0.0
 192.168.50.0/24    1        Direct     192.168.50.1    2.2.2.2         0.0.0.0

 Total Nets: 9
 Intra Area: 9  Inter Area: 0  ASE: 0  NSSA: 0

         OSPF Process 65534 with Router ID 128.1.138.137
                  Routing Tables

 Routing for Network
 Destination        Cost     Type       NextHop         AdvRouter       Area
 128.1.138.137/32   0        Direct     128.1.138.137   128.1.138.137   0.0.0.0

 Total Nets: 1
 Intra Area: 1  Inter Area: 0  ASE: 0  NSSA: 0
```

Gambar 25: Konfigurasi OSPF Huawei

1.5.3 3. Konfigurasi OSPF pada MikroTik



Gambar 26: Konfigurasi OSPF MikroTik

1.5.4 4. Verifikasi OSPF Neighbor

Gambar 27: Verifikasi OSPF Neighbor

1.5.5 5. Verifikasi Routing Table

Gambar 28: Verifikasi Routing Table

1.5.6 6. Verifikasi Koneksi End-to-End

```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.869 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.330 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.399 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.511 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=7.481 ms
```

```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.553 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.008 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.580 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.090 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.507 ms
```

```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.178 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.537 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.624 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=5.919 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.092 ms
```

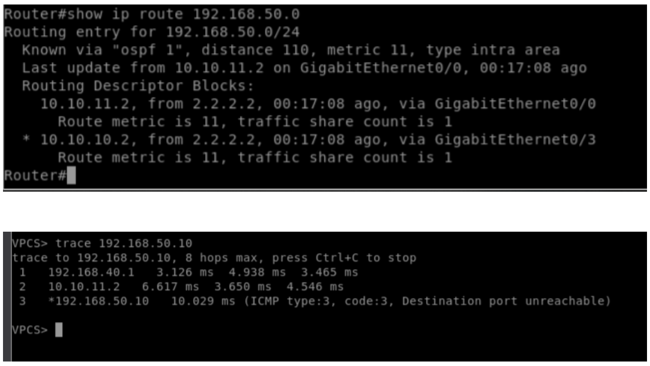
```
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.322 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.024 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.270 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.187 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=9.440 ms
```

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=13.408 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.324 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.915 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.182 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.026 ms

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=10.881 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=5.476 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=8.428 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=6.018 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=7.487 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=9.447 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=7.333 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=6.349 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=5.325 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=62 time=7.467 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.118 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=7.522 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=6.429 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=5.981 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=6.456 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=6.792 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=5.332 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=5.933 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=6.470 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=6.830 ms

VPCS>
```

Gambar 29: Verifikasi Koneksi End-to-End

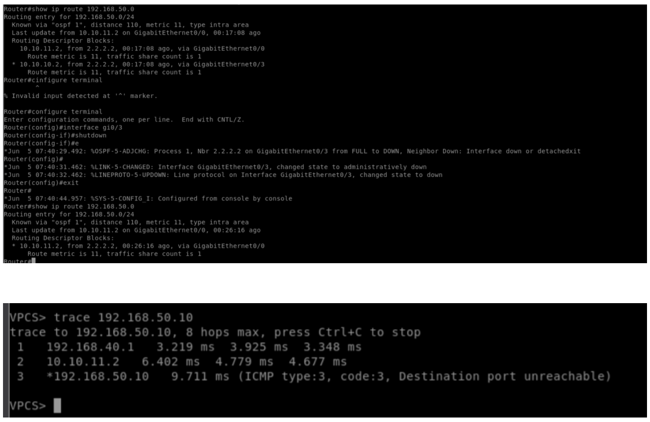
1.6 Praktikum 5 Pengujian Failover OSPF

1.6.1 1. Kondisi Normal: Dua Link Cisco-Huawei Aktif



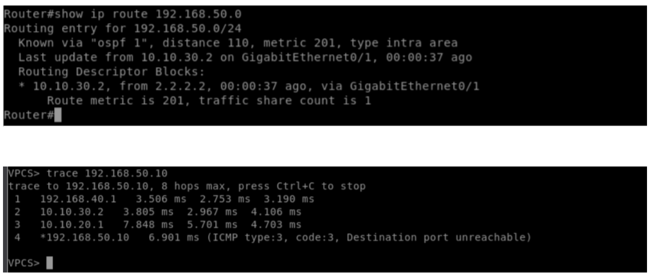
Gambar 30: Kondisi Normal

1.6.2 2. Kondisi Satu Link Cisco-Huawei Mati



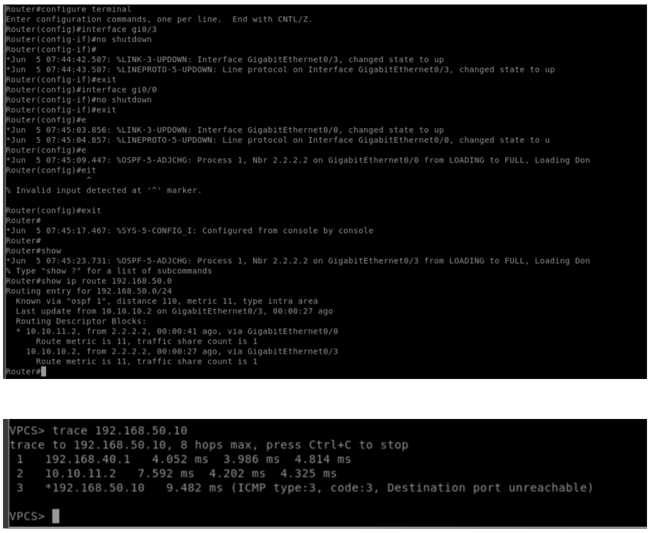
Gambar 31: Satu Link Cisco-Huawei Mati

1.6.3 3. Kondisi Dua Link Cisco-Huawei Mati



Gambar 32: Dua Link Cisco-Huawei Mati

1.6.4 4. Mengaktifkan Kembali Dua Link Cisco-Huawei



Gambar 33: Mengaktifkan Kembali Link Utama

2 Analisis Hasil Percobaan

Pada praktikum ini berhasil dilakukan konfigurasi VLAN, trunking, Router-on-a-Stick, DHCP Server, konfigurasi antar-router multivendor, implementasi OSPF, serta pengujian failover jaringan. VLAN berhasil memisahkan domain broadcast sesuai fungsi masing-masing divisi. Cisco Router berhasil menjalankan fungsi inter-VLAN routing sehingga perangkat pada subnet berbeda dapat saling berkomunikasi.

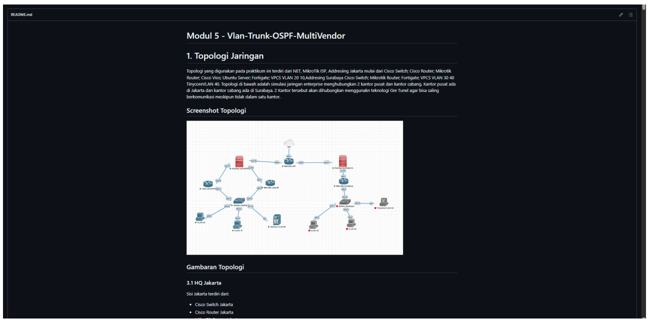
Konfigurasi DHCP pada VLAN 10 dan VLAN 20 berjalan dengan baik, dibuktikan dengan klien yang memperoleh alamat IP secara otomatis sesuai rentang yang telah ditentukan. Pada konfigurasi antar-router, seluruh perangkat Cisco, Huawei, dan MikroTik berhasil saling terhubung menggunakan alamat IP point-to-point yang sesuai.

Implementasi OSPF multivendor berhasil membentuk adjacency antara ketiga router. Routing table menunjukkan seluruh network berhasil dipelajari secara dinamis tanpa perlu menambahkan static route. Pengujian end-to-end menunjukkan seluruh VLAN dapat mengakses LAN Huawei dengan baik.

Pada pengujian failover, OSPF berhasil mendeteksi perubahan topologi jaringan secara otomatis. Saat satu link utama mati, trafik tetap menggunakan jalur utama yang tersisa. Ketika kedua link utama dimatikan, OSPF mengalihkan jalur komunikasi melalui MikroTik sebagai jalur cadangan. Setelah link utama diaktifkan kembali, trafik otomatis kembali menggunakan jalur dengan cost paling rendah. Hasil ini sesuai dengan teori kerja OSPF yang memilih jalur berdasarkan nilai cost dan mendukung redundansi jaringan.

3 Hasil Tugas Modul

Hasil pengerjaan tugas modul telah diunggah pada repository GitHub kelompok dalam folder `LA/README.md`. Dokumentasi hasil tugas modul ditampilkan pada gambar berikut.



Gambar 34: Hasil Tugas Modul

4 Kesimpulan

Berdasarkan seluruh rangkaian praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, trunking, Router-on-a-Stick, DHCP Server, routing antar-router multivendor, serta OSPF berhasil diterapkan dengan baik. Seluruh perangkat jaringan mampu berkomunikasi sesuai desain topologi yang telah dibuat. OSPF berhasil melakukan pertukaran informasi routing secara dinamis antar perangkat Cisco, Huawei, dan MikroTik.

Selain itu, pengujian failover membuktikan bahwa OSPF mampu menyediakan mekanisme redundansi jaringan dengan melakukan perpindahan jalur secara otomatis ketika jalur utama mengalami gangguan. Dengan demikian, seluruh tujuan praktikum berhasil dicapai dan hasil yang diperoleh sesuai dengan teori yang dipelajari.

5 Lampiran

5.1 Dokumentasi saat praktikum

Gambar 35: Dokumentasi Praktikum